

Question 1.

$$\sqrt{K_R} = \frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}}$$

$$= \frac{3.3 - 0.7 - 1.3}{1.3 - 0.6}$$

$$K_R = 3.45$$

$$\frac{K_n}{K_p} = 3.45$$

$$\frac{\mu_{n\text{ox}} \left(\frac{K'}{L}\right)_n}{\mu_{p\text{ox}} \left(\frac{W}{L}\right)_p} = 3.45$$

$$\frac{60 \times 10^{-6} \times 12}{25 \times 10^{-6} \times (W/L)_p} = 3.45$$

$$\underline{(W/L)_p = 8.35}$$

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{K_R}} (V_{DD} + V_{T0,p})}{1 + \sqrt{\frac{1}{K_R}}}$$

$$V_{T0,n} = 0.6 \pm (10\% \text{ of } 0.6) = 0.54 \text{ min} \\ = 0.66 \text{ max}$$

$$V_{T0,p} = -0.7 \pm (15\% \text{ of } 0.7) = -0.805 \text{ min} \\ = -0.595 \text{ max}$$

$$V_{th, \text{max}} = \frac{V_{T0,n \text{ max}} + \sqrt{\frac{1}{3.45}} (3.3 + V_{T0,p \text{ max}})}{1 + \sqrt{\frac{1}{3.45}}}$$

$$= \frac{0.66 + \sqrt{\frac{1}{3.45}} (3.3 - 0.595)}{1 + \sqrt{\frac{1}{3.45}}}$$

$$= \frac{2.1207}{1.54} = \underline{1.38 \text{ V}}$$

$$V_{th, \text{min}} = \frac{V_{T0,n \text{ min}} + \sqrt{\frac{1}{3.45}} (3.3 + V_{T0,p \text{ min}})}{1 + \sqrt{\frac{1}{3.45}}}$$

$$= \frac{0.54 + \sqrt{\frac{1}{3.45}} (3.3 - 0.805)}{1 + \sqrt{\frac{1}{3.45}}}$$

$$= \frac{1.8873}{1.54} = \underline{1.23 \text{ V}}$$

Question 2

Here $V_{T0,n} = |V_{T0,p}|$

For symmetric inverter

$V_{T0,n} = |V_{T0,p}|$ and $K_R = 1$

$$\frac{K_n}{K_p} = 1$$

$$\frac{\mu_n C_{ox} \left(\frac{W}{L}\right)_n}{\mu_p C_{ox} \left(\frac{W}{L}\right)_p} = 1$$

$$\frac{60 \times 12}{30 \times \left(\frac{W}{L}\right)_p} = 1$$

$$\therefore \left(\frac{W}{L}\right)_p = 24$$

$$\begin{aligned} V_{IL} &= \frac{1}{8} (3V_{DD} + 2V_{T0,n}) \\ &= \frac{1}{8} (3 \times 3.3 + 2 \times 0.6) \\ &= \underline{\underline{1.3875 \text{ V}}} \end{aligned}$$

$$\begin{aligned} V_{IH} &= \frac{1}{8} (5V_{DD} - 2V_{T0,n}) \\ &= \frac{1}{8} (5 \times 3.3 - 2 \times 0.6) \\ &= \underline{\underline{1.9125 \text{ V}}} \end{aligned}$$